DSCI-560
Lecture 6: Domain-Specific Examples

Data Science Professional Practicum

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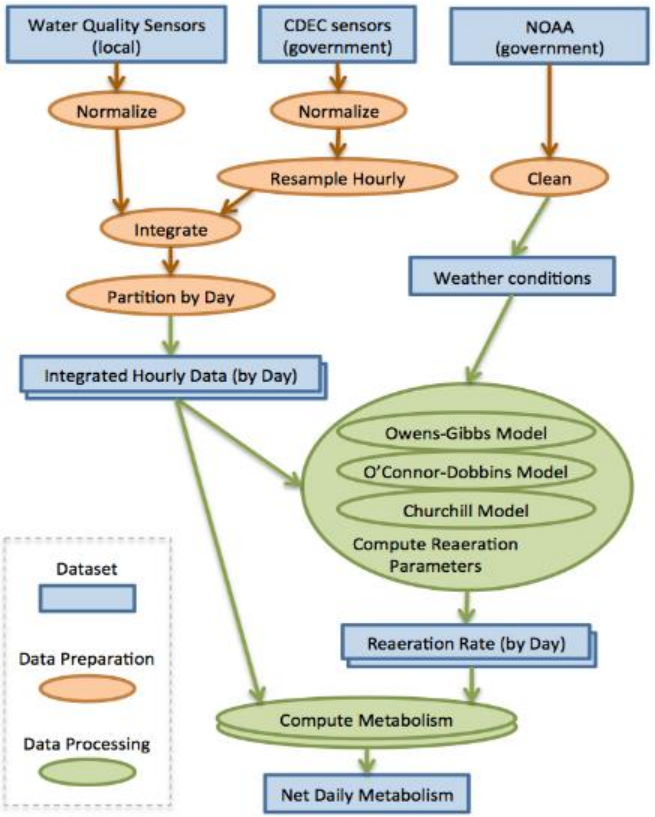
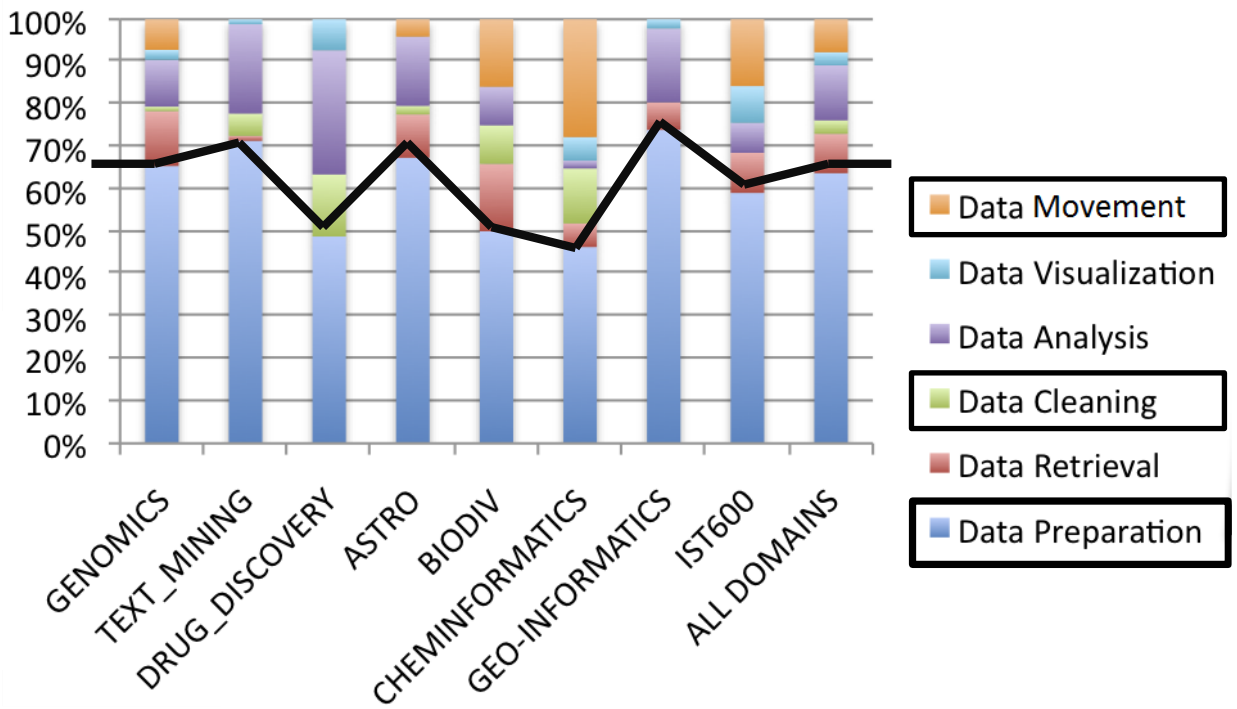
Data Analytics Workflow

- Data Preparation
  - Dataflow Plan
  - Dividing Large Loads
- Data Analytics
  - Merging the Results
  - Manual and Automated
- Scientific Conclusion
  - Hypothesis
  - Sufficient Experiments
  - Observe, then re-hypothesize
  - 1- modification, then repeat
  - Accurate Statistics
  - Empirical Evidence



Real Life: Data Prep IS 60-80%

- "Scientists and engineers spend more than 60% of their time just preparing the data for model input or data-model comparison" (NASA A40)



"Common Motifs in Scientific Workflows: An Empirical Analysis." Garijo, D.; Alper, P.; Belhajjame, K.; Corcho, O.; Gil, Y.; and Goble, C. Future Generation Computer Systems, 2013.

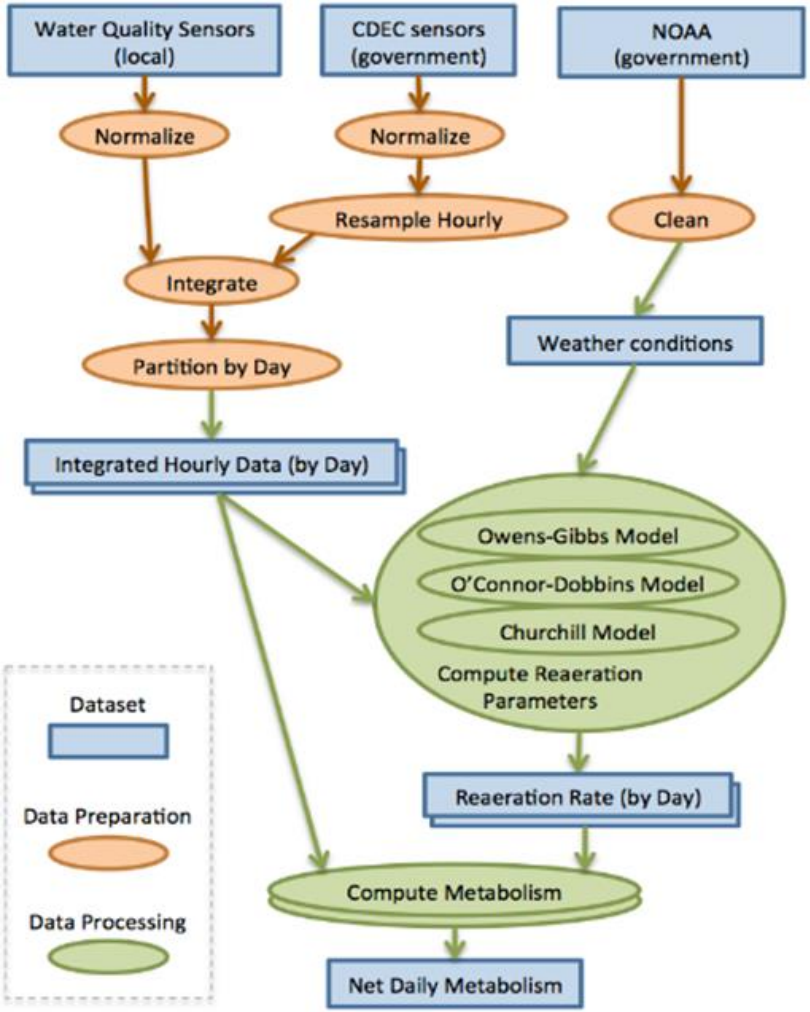
Accounting for Various Sources

- Data from Industry
  - Proprietary
  - Unlikely shared (possibly, share if anonymity)
  - Shared at a price
- Data from Academia
  - Open Source
  - Often Non-standard
  - Lack of reliable support
- Data from Government Organizations
  - Open Source
  - Variety of Standards and Format
  - Sporadic expertise
  - Lack of reliable support

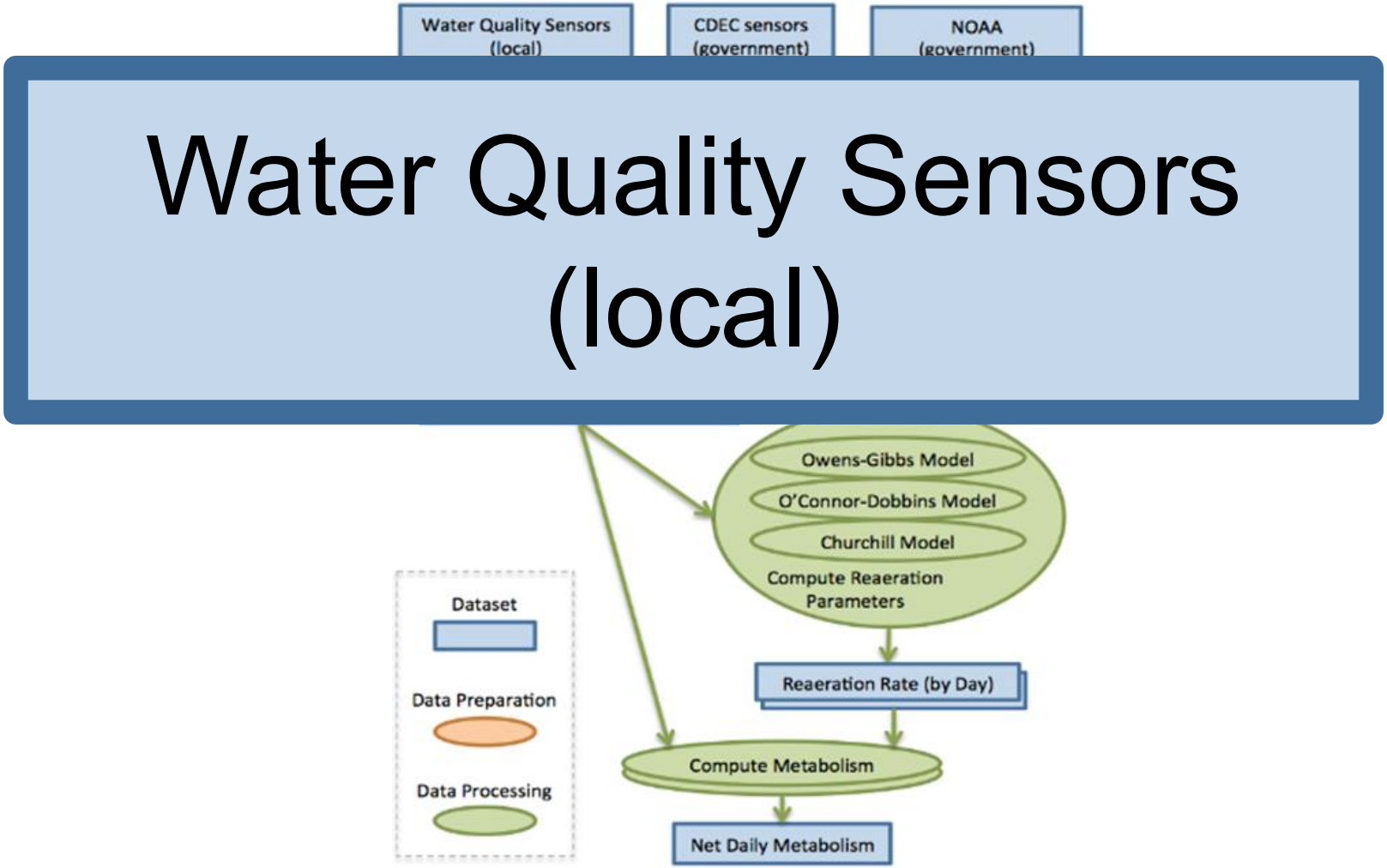
Data for Your Problem

- Data Availability
  - Sensor Deployments
  - Data Exchange
- Data Quality
  - Sensor Capability and Limitations
  - Accuracy and Sampling Rates
  - Post Processing Impacts
- Data Coverage
  - Sufficient to Solve the Problem?

Example: Water Quality Prediction



Data Preparation: Sensors



Water Quality Sensors

Water Quality Sensors

Sensor Specifications

Sensor Summary

Sensors	Expected Lifetime*	Recommended Calibration Frequency	Pressure Rating - PSI	Usable Depth m | ft		Operational Temperature Range
pH/ORP	2 years or greater**	10 to 12 weeks**	350	250	820	- 5 to 50° C
RDO	2 years or greater	12 months**	350	250	820	- 5 to 50° C
Conductivity	2 years or greater	User calibration only if needed	350	250	820	- 5 to 50° C
Temperature	2 years or greater	NA	350	250	820	- 5 to 50° C
Turbidity	2 years or greater	User calibration only if needed	350	250	820	- 5 to 50° C
Pressure	2 years or greater	User calibration only if needed	12.8 14.2 42.7 108 142 285 350	9 10 30 76 100 200 250	30 33 100 250 328 650 820	- 5 to 50° C
Barometric	2 years or greater	User calibration only if needed	NA	NA	NA	- 5 to 50° C

...

Data Sampling Rate

Environmental rating	IP68 with all sensors and cable attached. IP67 with sensors removed or cable detached
Max pressure rating	Up to 350 PSI Ammonium/Nitrate up to 30 PSI
Communication	RS485/MODBUS, SDI-12, Bluetooth®
Reading rate	1 reading every 2 seconds for one parameter, no wipe
LCD screen	Integrated display shows status of sonde, sensor ports, power voltage and connectivity.
External power voltage External power current (Dependent on display/wiping)	8-36 VDC (required for normal operation) Sleep: < 0.2 mA typical Measurement: 40 mA typical, 75 mA Max
Cable	Vented or non-vented polyurethane or vented Tefzel®
Hex screw driver	0.050 in. (1.3 mm)
Software Interface	Android/iOS: VuSitu Mobile App (see app store for OS requirements) Data Services: HydroVu

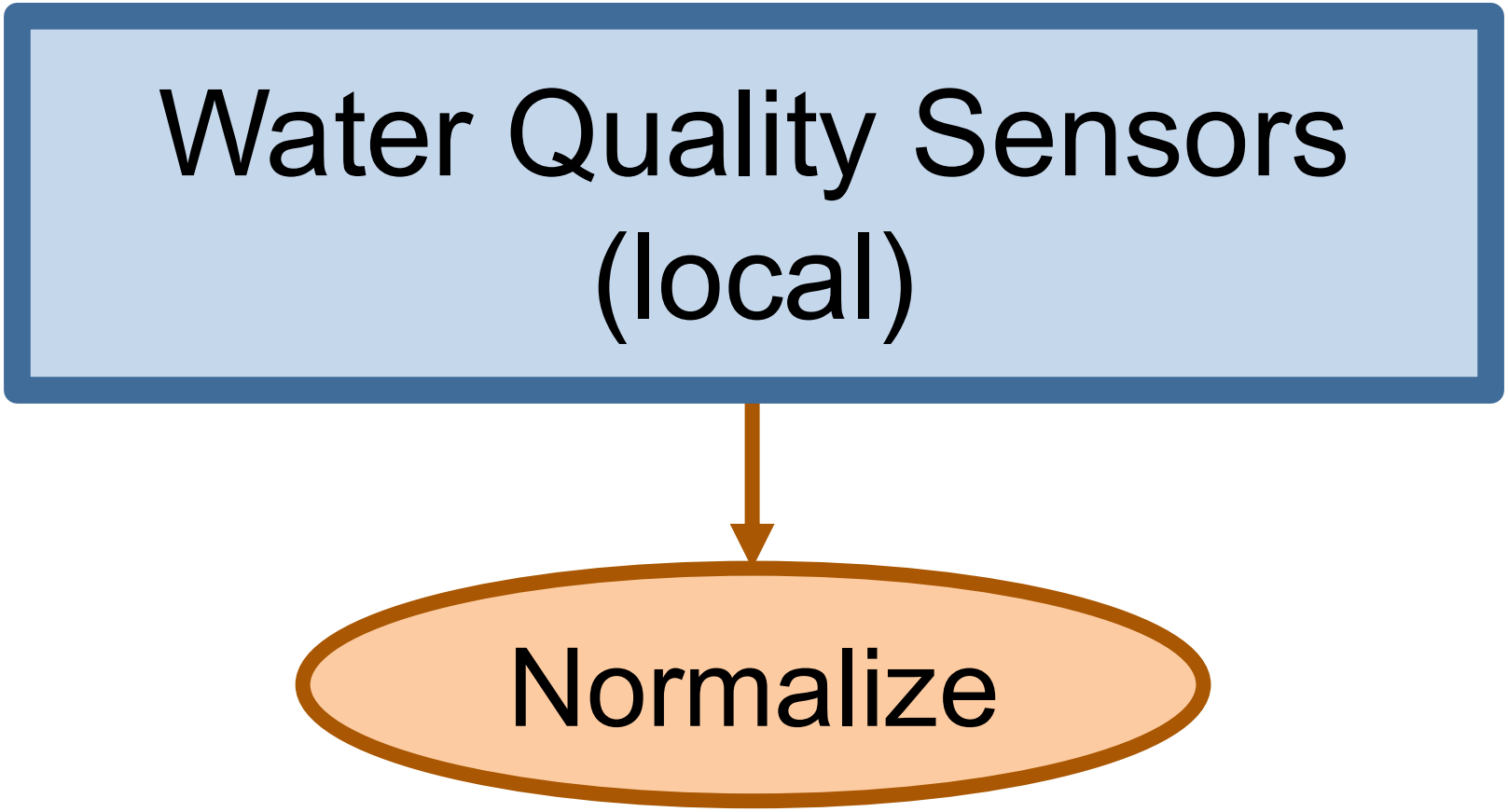
Data Format

READING RATES	1 reading every 2 seconds
DATA LOGGING	50 logs (defined, scheduled to run, or stored)
LOGGING MODES	Linear, Linear Average, Event
LOGGING RATE	1 minute to 99 hours

- Sampling Rate
  - Seconds, Minutes, Hours
  - Example: 0.5 Hz logging every minutes-hours
- Bit Width
  - 10-12 bits typical for ADC
  - 8, 16, 32 bits per data
- Range
  - Various ranges for different data
  - Example: -5 to 50 degrees at 0.1
- Data Type
  - Signed/Unsigned
  - Floating-point/Fixed-point
  - IEEE?

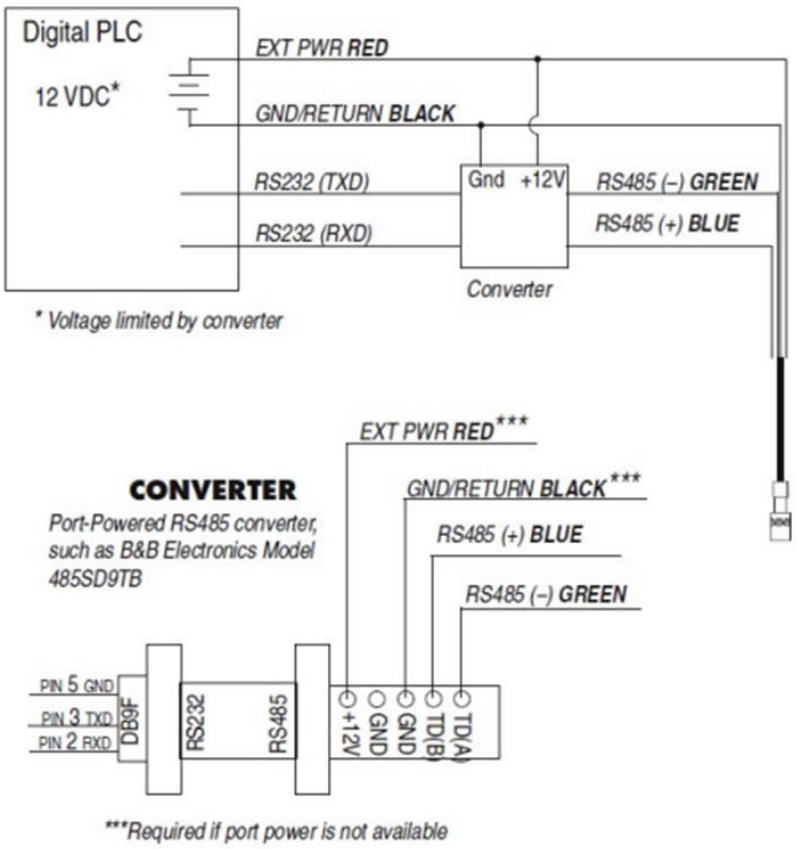
SENSOR	ACCURACY	RANGE
TEMPERATURE[5]	± 0.1° C	-5 to 50° C (23 to 122° F)
BAROMETRIC PRESSURE	± 1.0 mbars	300 to 1,100 mbar
pH[6]	±0.1 pH unit or better	0 to 14 pH units
ORP[7]	±5 mV	±1,400 mV
CONDUCTIVITY[8]	±0.5% of reading plus 1 µS/cm from 0 to 100,000 µS/cm; ±1.0% of reading from 100,000 to 200,000 µS/cm; ±2.0% of reading from 200,000 to 350,000 µS/cm	0 to 350,000 µS/cm

Data Preparation: Sensor Data

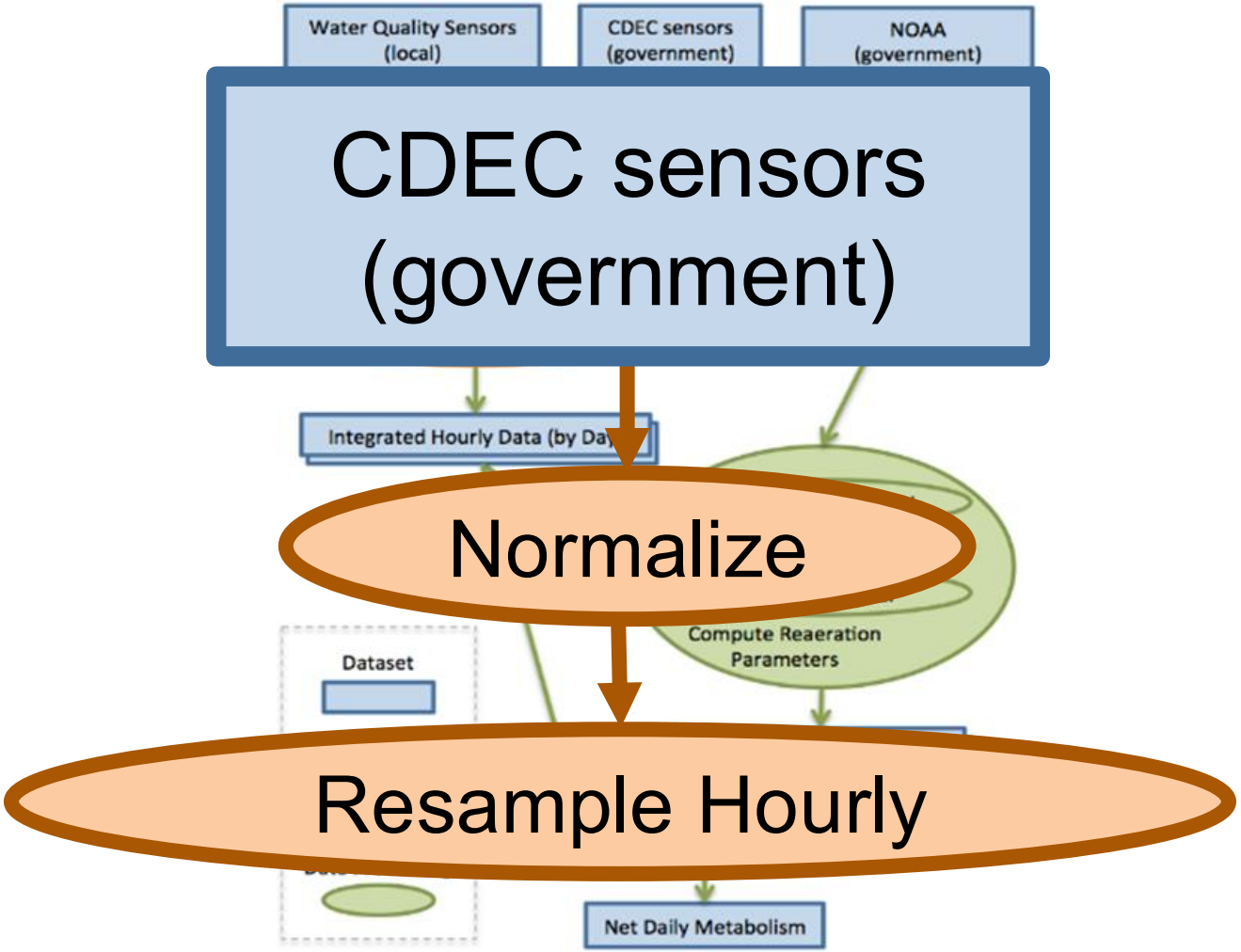


Programmable Logic Controller

- Three Stages
  - Inputs
  - Program Execution
  - Outputs
- Standard Interface for Input
  - Variety types
  - Example: RS232
- Program
  - Scaling and Normalization
  - Data Conversions
  - Response to In-situ Data
  - Other Custom Operations
- Standard Output Interface
  - Usually more complex interface
  - Ethernet, WiFi, etc.



Data Preparation: Sensors



California Data Exchange Center

Share: About Contact Current Conditions Settings

CALIFORNIA DATA EXCHANGE CENTER
CALIFORNIA DEPARTMENT OF
WATER RESOURCES

Home Query Tools Precipitation River Forecast River Stages Reservoirs Snow Stations Weather

MALIBU *(MLB)*

Elevation: 1575' · VENTURA LA COASTAL basin · Operator: Los Angeles County Fire Dept
Query executed Tuesday at 16:29:52

Provisional data, subject to change.
Select a sensor type for a plot of data.

Hourly Data

Earlier

DATE / TIME PDT	RAIN INCHES	TEMP DEG F	REL HUM %	FUEL MS %	FUEL TP DEG F	WIND SP MPH	WIND DR DEG	PEAK WS MPH	PEAK WD DEG	SOLAR R W/M^2	BAT VOL VOLTS
09/05/2023 05:00	43.46	60	98	21	58	0	38	5	102	0.00	12.5
09/05/2023 06:00	43.46	59	99	22	59	3	56	6	85	3.00	12.5
09/05/2023 07:00	43.46	61	98	22	61	4	110	8	118	33.00	12.6
09/05/2023 08:00	43.46	63	92	21	65	2	118	6	65	100.00	13.0
09/05/2023 09:00	43.46	63	91	20	66	3	142	7	109	144.00	13.4
09/05/2023 10:00	43.46	63	92	17	70	5	164	8	161	271.00	13.8
09/05/2023 11:00	43.46	66	83	14	74	4	150	6	160	393.00	13.8
09/05/2023 12:00	43.46	68	81	13	77	3	199	7	154	392.00	13.9
09/05/2023 13:00	43.46	69	76	13	82	4	161	7	109	529.00	14.1
09/05/2023 14:00	43.46	69	78	11	79	4	157	10	129	580.00	13.4
09/05/2023 15:00	43.46	68	80	11	78	3	144	8	166	477.00	13.9
09/05/2023 16:00	--	--	--	--	--	--	--	--	--	--	--

Showing 1 to 12 of 12 entries

Copy Excel CSV Print PDF

CDEC Data in CSV format

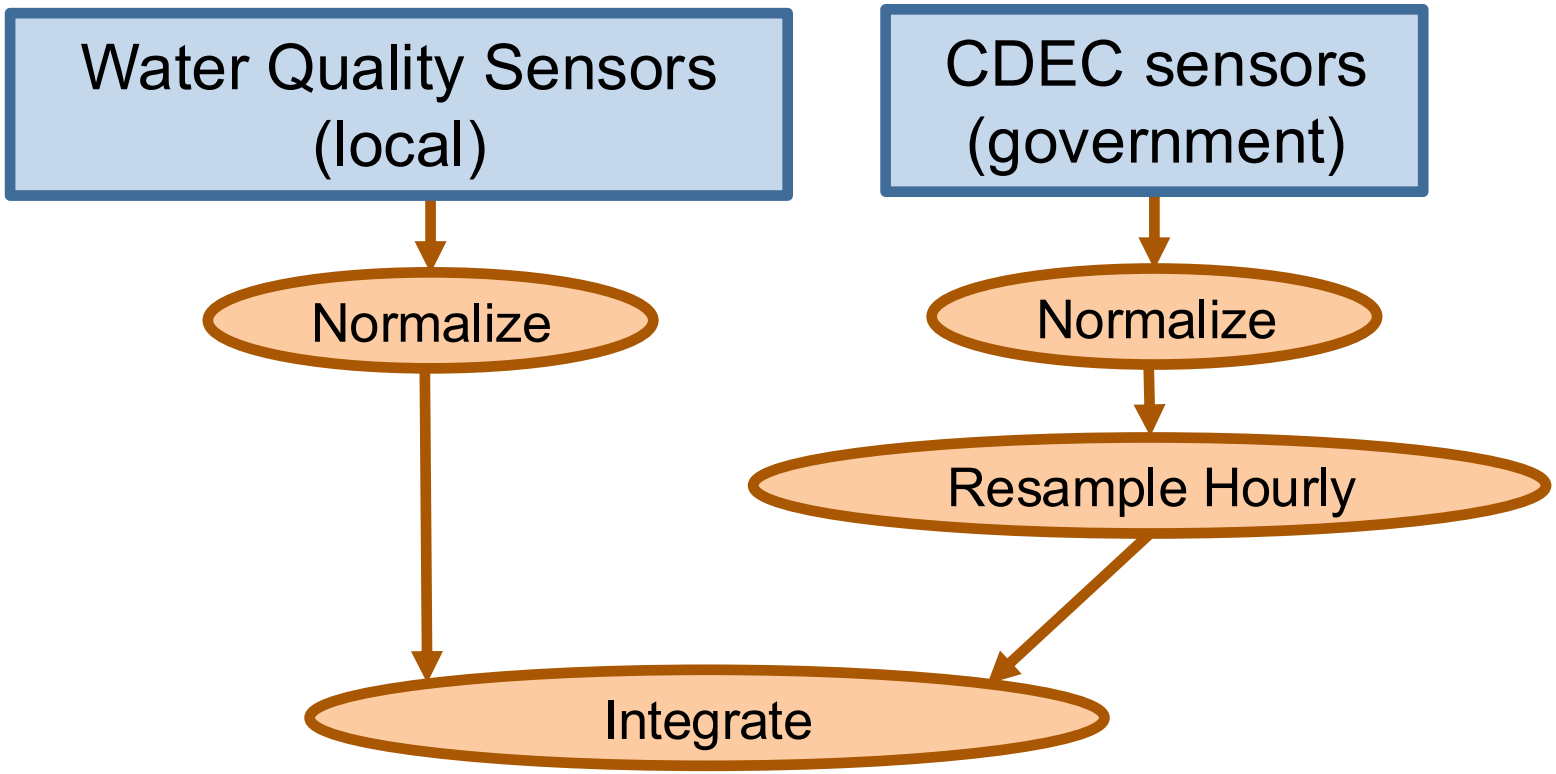
MALIBU (MLB)

File Edit View

```
"DATE / TIME PDT","RAIN INCHES","TEMP DEG F","REL HUM   %","FUEL MS   %","FUEL TP DEG
F","WIND SP MPH","WIND DR DEG","PEAK WS MPH","PEAK WD DEG","SOLAR R W/M^2","BAT VOL
VOLTS"
"09/05/2023 19:00   ","43.46","63","88","14","63","1","228","3","189","4.00","13.2"
"09/05/2023 20:00   ","43.46","64","86","15","64","0","53","4","167","0.00","13.1"
"09/05/2023 21:00   ","43.46","63","88","16","63","2","178","4","191","0.00","13.0"
"09/05/2023 22:00   ","43.46","63","90","17","63","3","288","4","259","0.00","12.9"
"09/05/2023 23:00   ","43.46","62","91","18","61","1","150","3","301","0.00","12.8"
"09/06/2023 00:00   ","43.46","61","94","19","59","0","57","2","168","0.00","12.7"
"09/06/2023 01:00   ","43.46","62","91","20","60","0","174","3","250","0.00","12.6"
"09/06/2023 02:00   ","43.46","62","88","20","60","0","354","2","353","0.00","12.6"
"09/06/2023 03:00   ","43.46","62","88","21","61","0","15","2","318","0.00","12.6"
"09/06/2023 04:00   ","43.46","62","87","20","62","1","132","3","24","0.00","12.6"
"09/06/2023 05:00   ","43.46","62","86","20","62","2","275","4","265","0.00","12.5"
"09/06/2023 06:00   ","--","--","--","--","--","--","--","--","--","--","--"
```

Ln 13, Col 77 | 100% | Windows (CRLF) | UTF-8

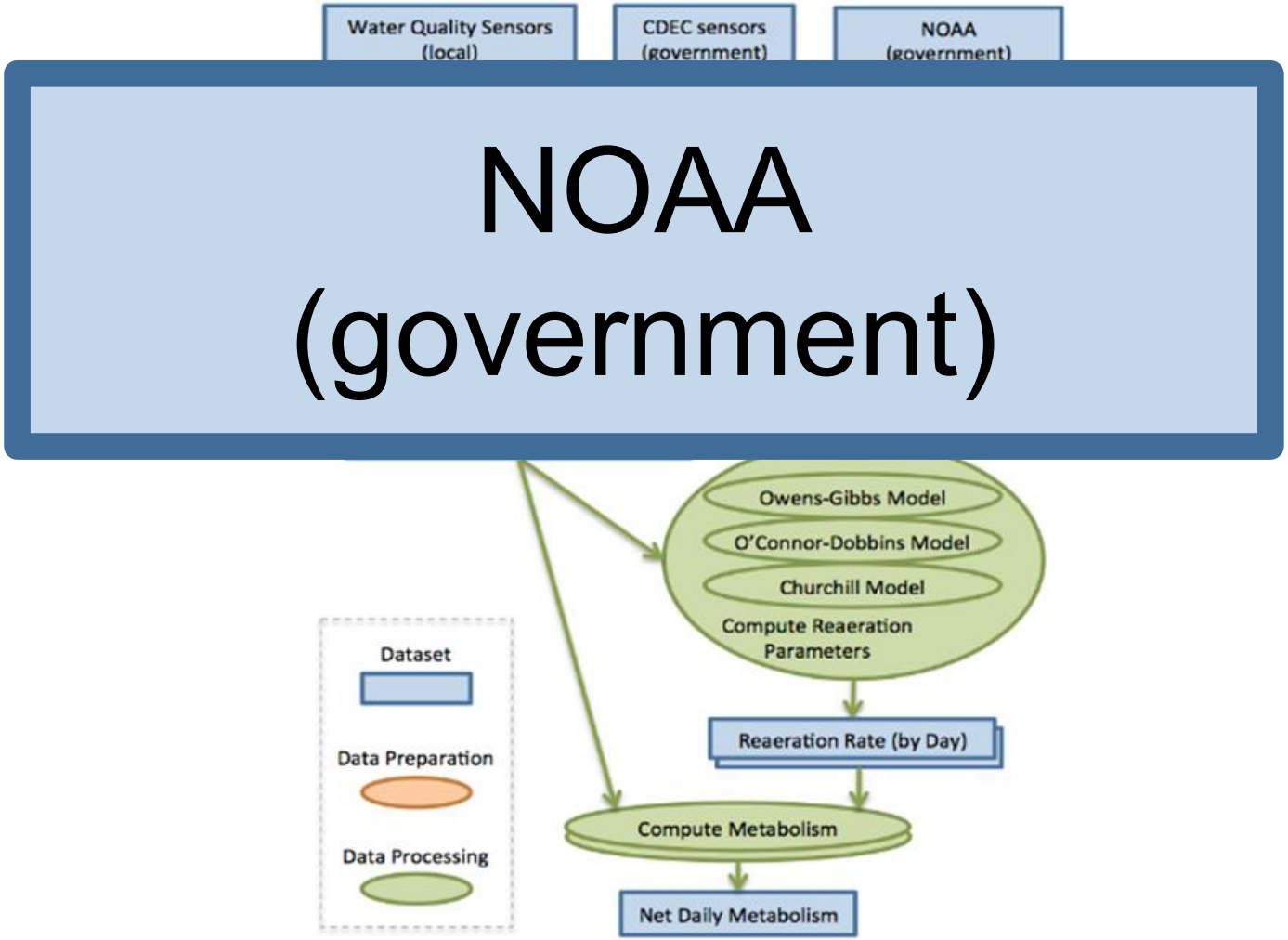
Data Integration



Integrating Sensor Data

- Matching Sampling Rates
  - 1 Sample every 2 seconds (local sensors)
  - 1 Sample every hour (sensor station data)
  - Example decisions: interpolate or average
  - Depends on the backend analysis
- Prepare Based on Application
  - Scaling and normalization
  - Weights associated with data

Data Preparation: Sensors



National Oceanic and Atmospheric Administration

NATIONAL WEATHER SERVICE
NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION

HOME | FORECAST | PAST WEATHER | SAFETY | INFORMATION | EDUCATION | NEWS | SEARCH | ABOUT

Local forecast by "City, St" or ZIP code
Enter location ... Go
Location Help

Heat Wave Continues in the East and South; Watching Tropical Storm Lee in the Atlantic

A heat wave will persist in the Mid-Atlantic and Northeast through Thursday, including in the National Capital Region, and linger all week in Texas and neighboring states. Tropical Storm Lee is forecast to become a major hurricane over the western tropical Atlantic by this weekend and could bring impacts to the Leeward Islands by that time. Read More >

Customize Your Weather.gov

City, ST

Enter Your City, ST or ZIP Code

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API Web Service

Weather.gov > Documentation > API Web Service

Documentation
National Headquarters

Services | Technical Bulletins

Overview | Examples | Updates | Specification

Overview

The National Weather Service (NWS) API allows developers access to critical forecasts, alerts, and observations, along with other weather data. The API was designed with a cache-friendly approach that expires content based upon the information life cycle. The API is based upon of JSON-LD to promote machine data discovery.

The API is located at: https://api.weather.gov

Operational issues should be reported to nco.ops@noaa.gov.

General use questions can be asked on the API github site.

Application Programming Interface

API Web Service

Weather.gov > Documentation > API Web Service

Documentation
National Headquarters

Services Technical Bulletins

Overview | **Examples** | Updates | Specification

Examples of using the API

The API uses linked data to allow applications to discover content. Similar to a web site that provides HTML links to help users navigate to each page, linked data helps applications navigate to each endpoint. You may also review the OPEN API specification on the "Specification" tab on this page, or directly using the specification endpoint (that is also used to create the tab presentation): https://api.weather.gov/openapi.json.

How do I get the forecast?

Forecasts are created at each NWS Weather Forecast Office (WFO) on their own grid definition, at a resolution of about 2.5km x 2.5km. The API endpoint for the 12h forecast periods at a specific grid location is formatted as:

```
https://api.weather.gov/gridpoints/{office}/{gridX},{gridY}/forecast
```

For example: https://api.weather.gov/gridpoints/TOP/31,80/forecast

To obtain the grid forecast for a point location, use the /points endpoint to retrieve the current grid forecast endpoint by coordinates:

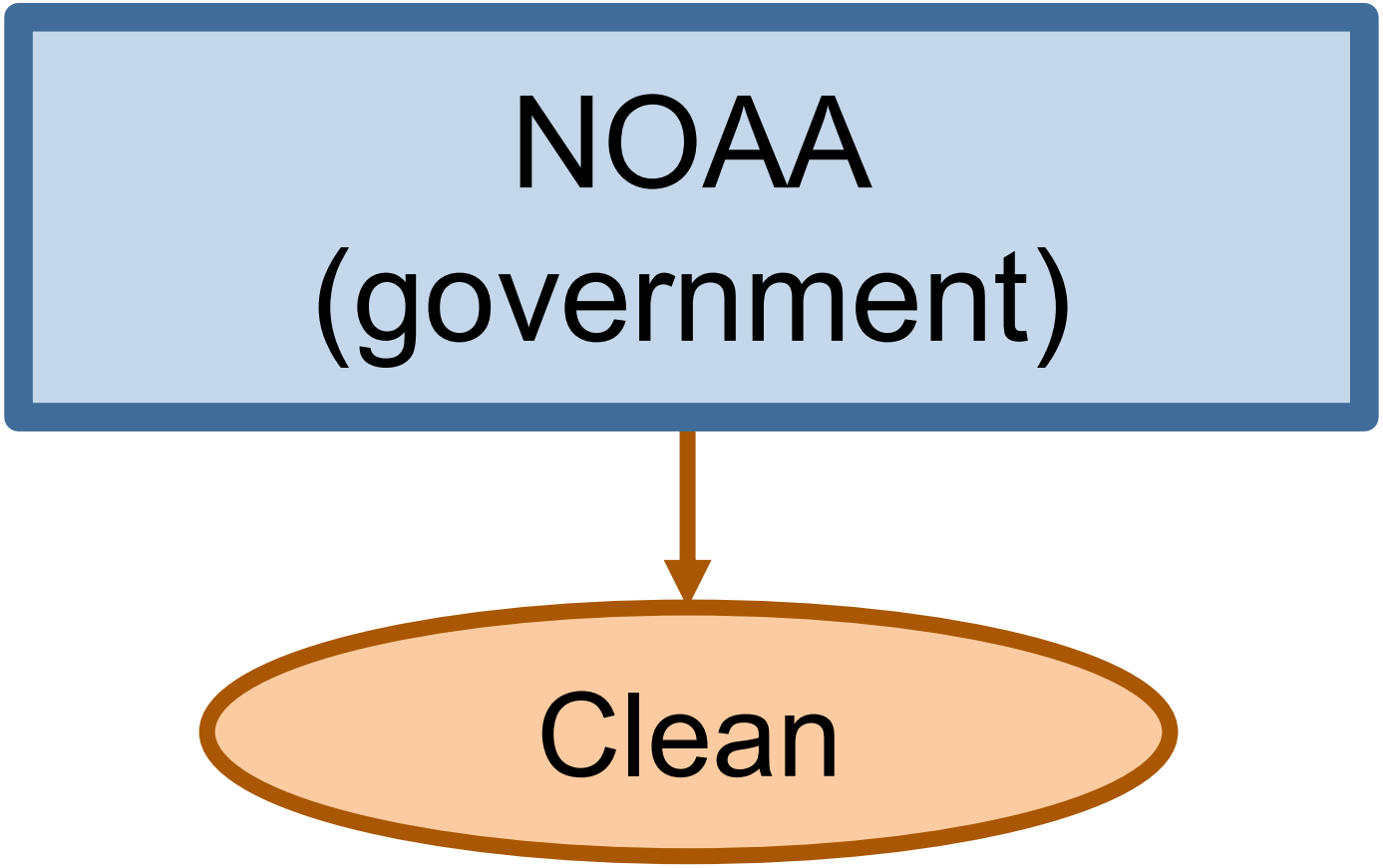
```
https://api.weather.gov/points/{latitude},{longitude}
```

JSON Output Example via Web

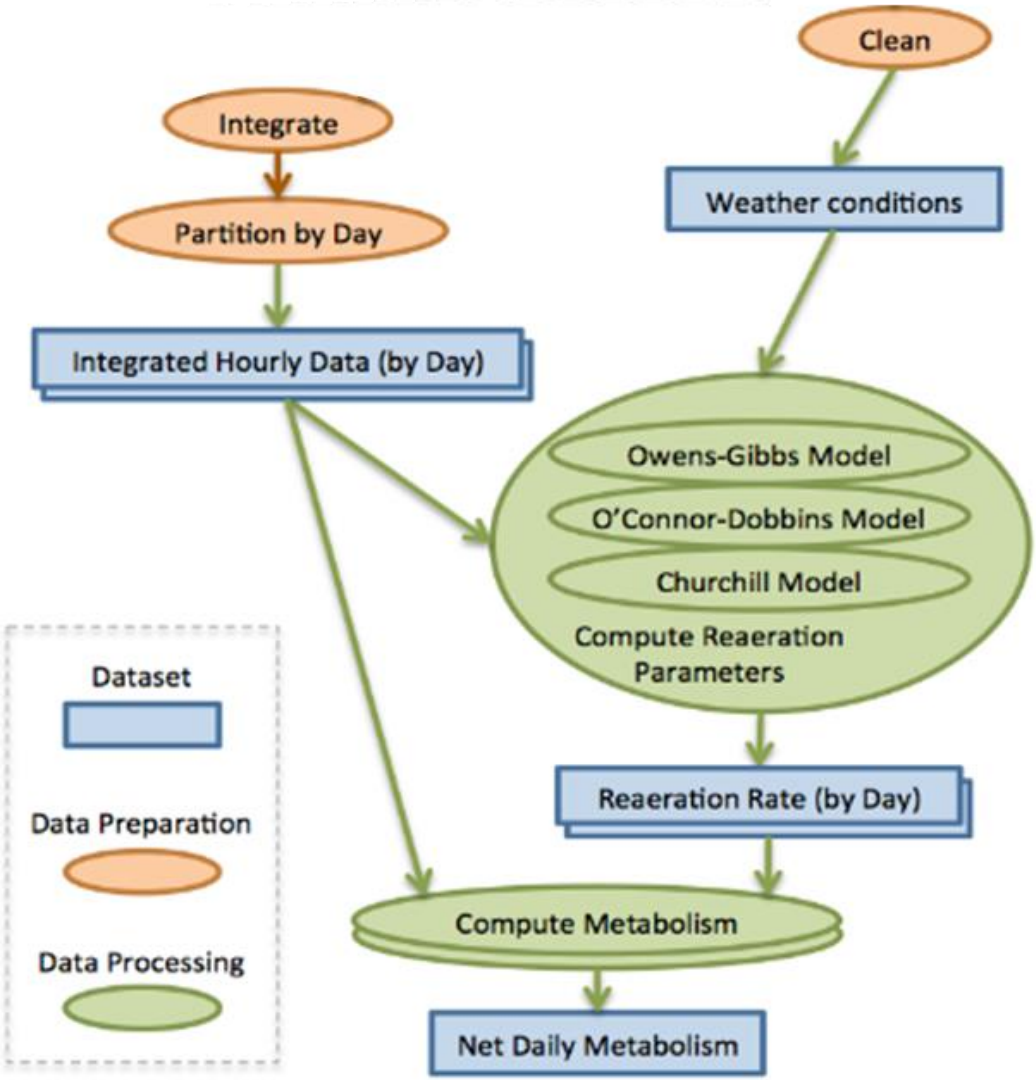
- https://api.weather.gov/points/39.7456,-97.0892


```
"properties": {
    "@id": "https://api.weather.gov/points/39.7456,-97.0892",
    "@type": "wx:Point",
    "cwa": "TOP",
    "forecastOffice": "https://api.weather.gov/offices/TOP",
    "gridId": "TOP",
    "gridX": 32,
    "gridY": 81,
    "forecast": "https://api.weather.gov/gridpoints/TOP/32,81/forecast",
    "forecastHourly": "https://api.weather.gov/gridpoints/TOP/32,81/forecast/hourly",
    "forecastGridData": "https://api.weather.gov/gridpoints/TOP/32,81",
    "observationStations": "https://api.weather.gov/gridpoints/TOP/32,81/stations",
    "relativeLocation": {
        "type": "Feature",
        "geometry": {
            "type": "Point",
            "coordinates": [
                -97.086661000000007,
                39.679375999999998
            ]
        },
        "properties": {
            "city": "Linn",
            "state": "KS",
            "distance": {
                "unitCode": "wmoUnit:m",
                "value": 7366.9851976443997
            },
            "bearing": {
                "unitCode": "wmoUnit:degree_(angle)",
                "value": 358
            }
        }
    },
    "forecastZone": "https://api.weather.gov/zones/forecast/KSZ009",
    "county": "https://api.weather.gov/zones/county/KSC201",
    "fireWeatherZone": "https://api.weather.gov/zones/fire/KSZ009",
    "timeZone": "America/Chicago",
    "radarStation": "KTWX"
}
```

Data Preparation: Sensors



Data Analysis Backend



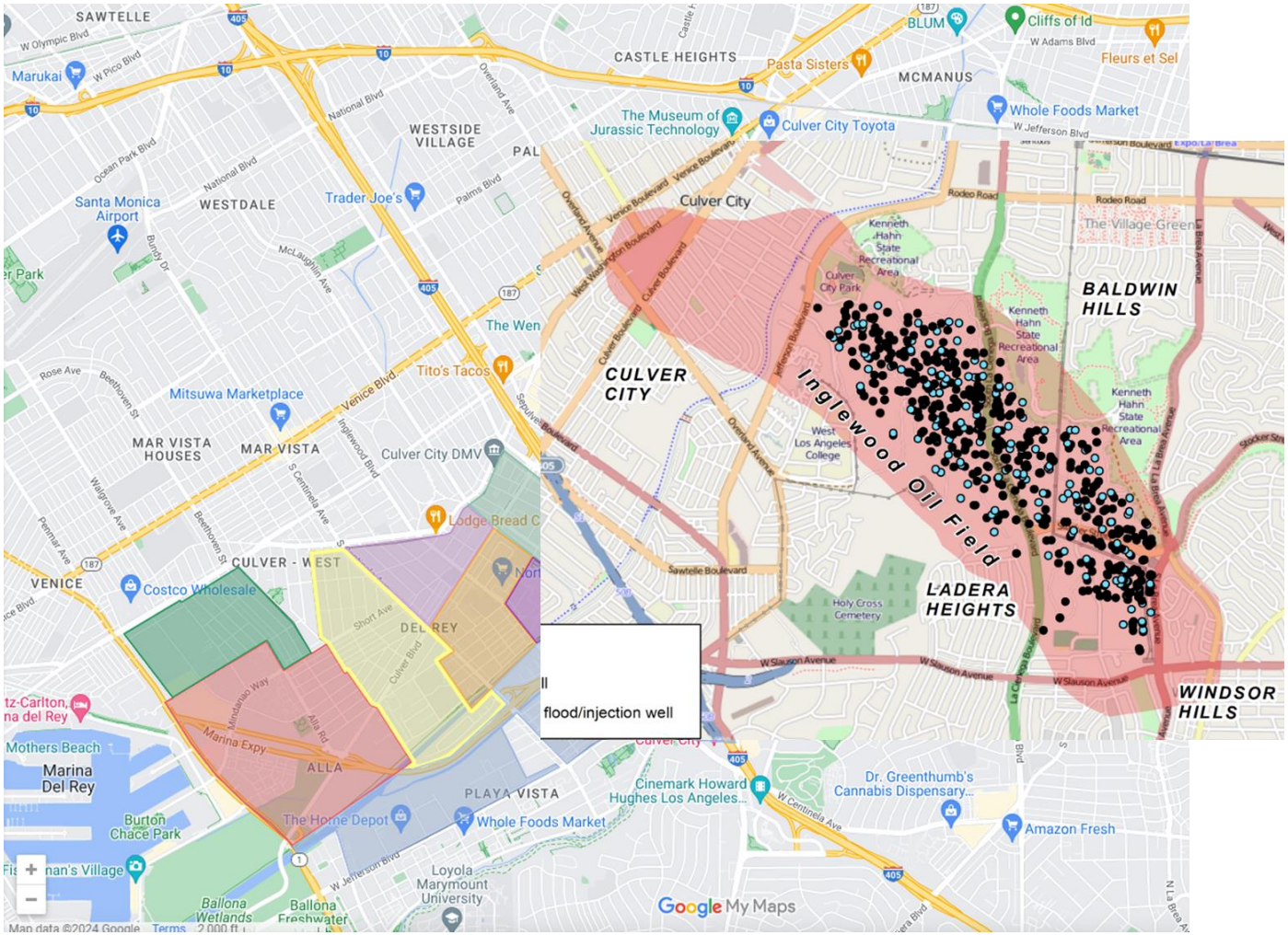
Leveraging Expertise

- Domain Knowledge
  - Scientific Models Developed over the Years
  - Model Refinements with Real Data
- Development of a New Model
  - Data Selection and Integration
  - Application of Scientific Methodology
- Expertise is Critically Important
  - Collaborate with the Experts
  - The Best Option: Learn to Become One

Methane from Oil Wells

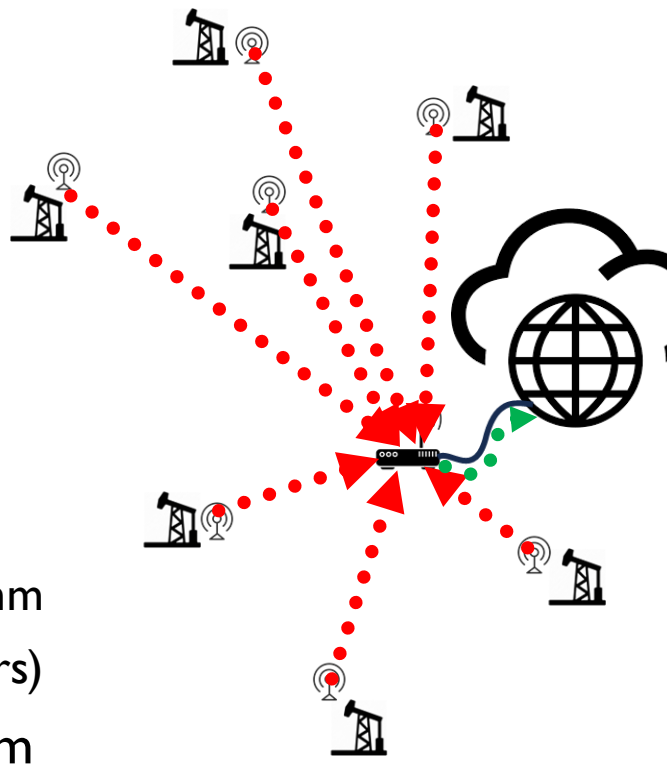
- U.S. EPA Definition
  - No production and not plugged
  - No production without operator: orphaned
  - No production and plugged
- Known Problem: Leaks
  - Methane Emissions (Air, Climate, Ecosystem)
  - Water Contamination (Surface and Groundwater)
  - Other Toxins like arsenic, benzene, and hydrogen sulfide
- Official Records
  - ~4.6 million Oil and Gas Wells
  - **~1.9 million Active**
  - **~2.7 million Abandoned (~58%)**
  - **~100k-1million Orphaned ?!?**
- A Possible Solution: Well Monitoring
  - Challenges like power, communication, maintenance costs

Methane Monitoring in LA



Objective: Methane Leak Monitoring

- Wireless Industrial Internet of Things
  - Low-Cost Modular and Scalable Solution
  - Low-Cost Installation and Deployment
  - Long Range Communication Capability
  - High-Sensitivity Methane Sensor
- Self-Sustaining Long-term Solution
  - Scalable Ambient Energy Harvesting
  - Ultra Low-Power IoT Platform
  - Energy Aware System Operation Algorithm
  - Extended Life Energy Storage (20-30 years)
- Data Analysis, Display, and Alert System
  - Automated Data Filtering/Cleaning and Storage
  - Data Analytics-based Models for Methane Leak



Approach: Design-Test-Repeat

- Application-Specific IIoT Design
  - Starting with Industrial Standard IIoT Platform
  - Integration of Rechargeable Power Storage
  - Integration of Energy Harvester
  - Integration of Methane Sensor
  - Integration of Wind Sensor Module
  - Design-Test-Repeat Cycle
- Cloud Data Storage and Analytics
  - Real-time data forwarding, collection, and display
  - Physics and data-driven modeling and prediction
- Experiments
  - Methane Leak Emulation Instruments
  - Outdoors Experiments

Sensors: Methane Sensor

NevadaNano Methane Sensor

PERFORMANCE

Range*	50 - 1,000,000 ppm
Resolution	1 ppm
Calibration	Factory calibrated
Accuracy *(Typical, at 20C, 50 %RH)*	± 10% error (> 300 ppm)

**Onboard sensor algorithms utilize a smart, variable lower detection limit, updated every sensor cycle, based on "live" sensor conditions.*

ENVIRONMENTAL OPERATING RANGE

Temperature	−40 to 75 °C
Humidity	0 to 100 %RH
Pressure	80 to 120 kPa

FM APPROVED C US

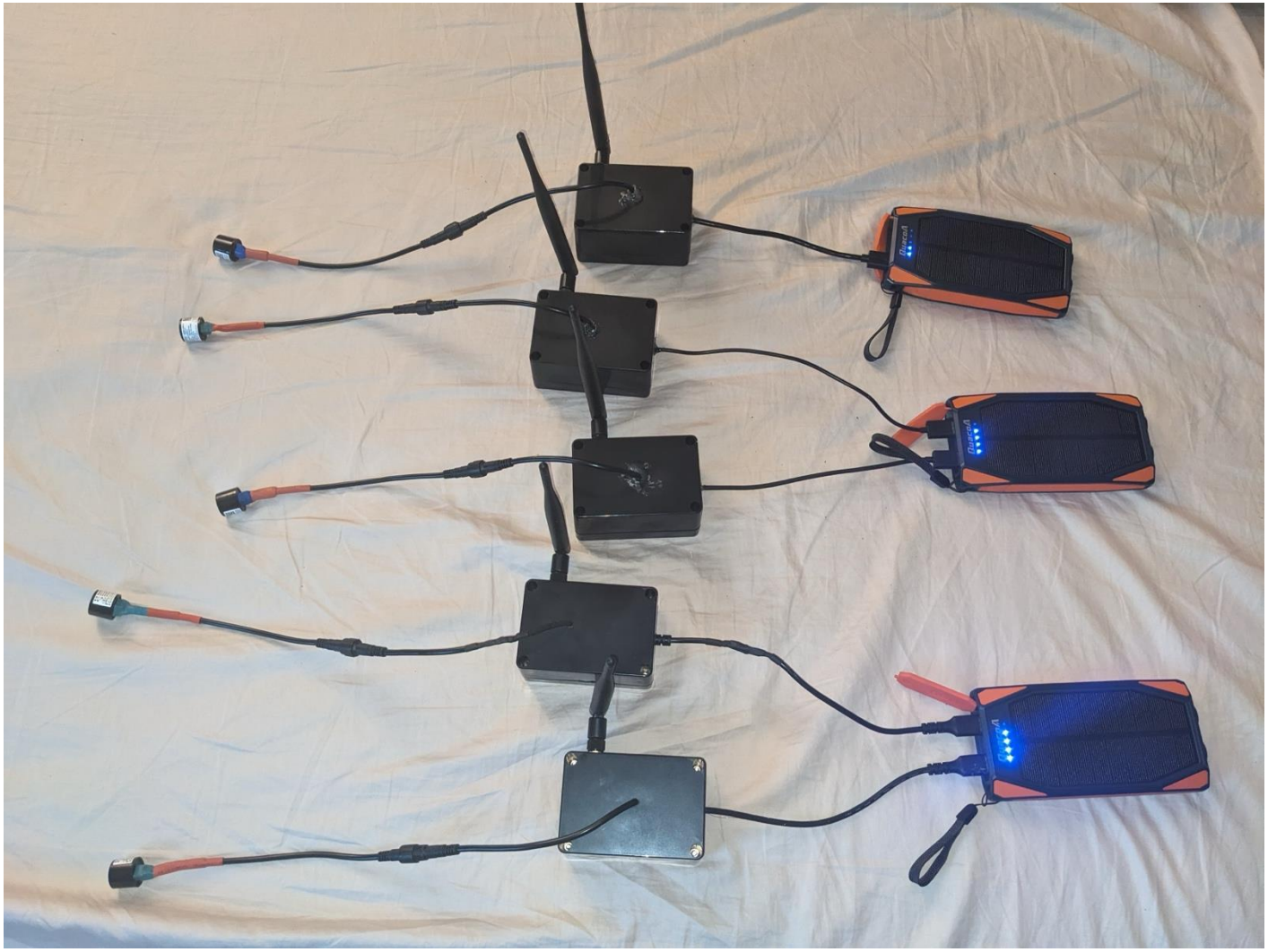
IECEx

ATEX UKEX

FEATURES

- Built-in environmental compensation
- Inherently poison-resistant
- No calibration required
- 5+ year lifetime
- Low power — 29 mW average
- Intrinsically safe (IS) certified
- Built-in self-test for fail-safe operation

5 Working Units



Canopy + IIoT - Vent

Data Collection and Prep

iiot_v4_sleep

File Edit View

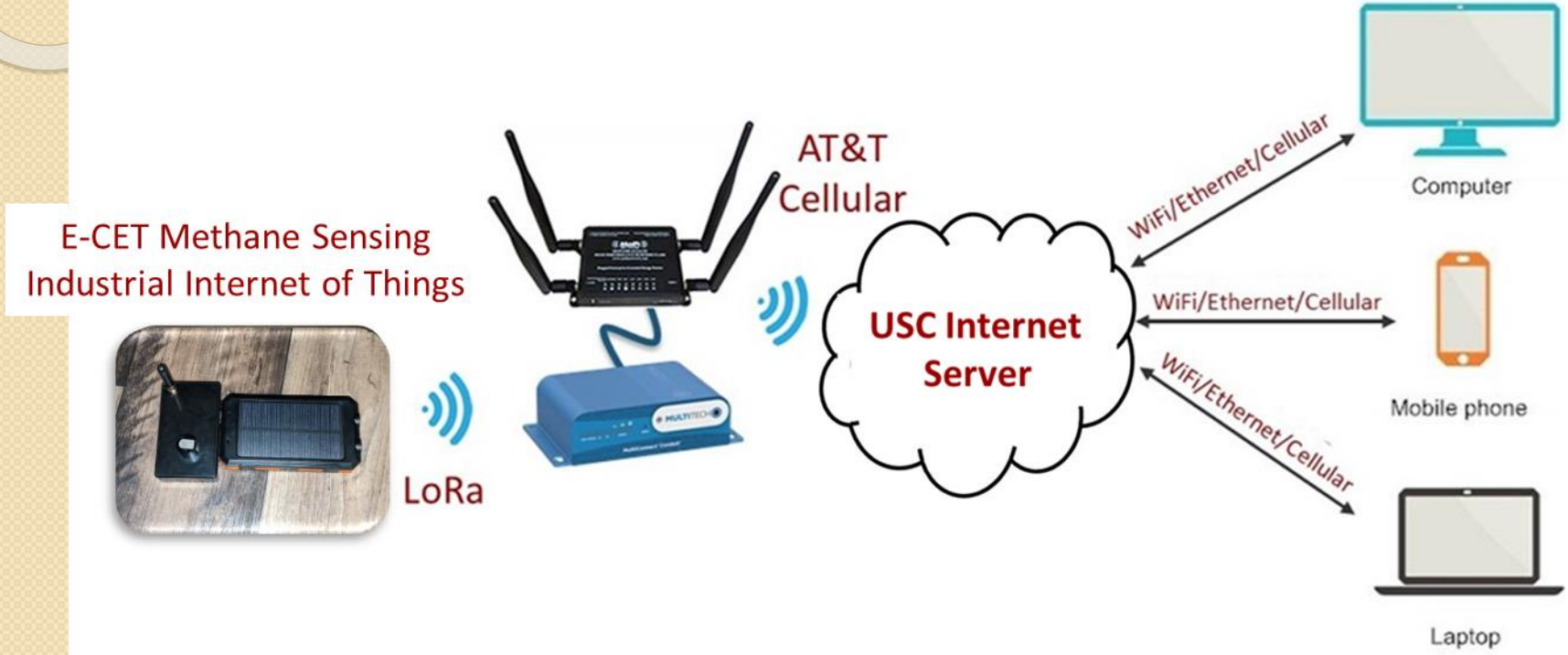
```
// Read data from Sensor
bool sensor_read(std::vector<uint8_t> *tx_data) {
    int tries = 5;
    bool success = false;
    int rcnt;
    uint8_t ccycle;
```

```
if ((read_buf[0]==0x01)&&(read_buf[1]==0)&&(ccycle!=pcycle)) {
    logInfo("[DONE] New data from Sensor.");
    pcycle=ccycle;
    success = true;
    tx_data[0].push_back('x');
    for (int j=6; j<rcnt; j=j+4) for (int i=j+3; i>=j; i--) {
        tx_data[0].push_back(read_buf[i]);
    }
} else {
    logInfo("[WAIT] Duplicate data from Sensor.");
    //ThisThread::sleep_for(500);
    dot_sleep(1);
}
```

```
logInfo("[WAIT] No response from Sensor.");
//ThisThread::sleep_for(500);
```

Ln 1, Col 1 | 100% | Windows (CRLF) | UTF-8

End-to-End System



Data Communication Limits

iiot_v4_sleep

File Edit View

```
        if (! boot) {// Attempt to read from the sensor
            if (! sensor_read(&tx_data)) {
                logInfo("[FAIL] Sensor failed to produce data 5 Times.");
                send_err_packet(0xff);
            }

            if (! boot) {
                // Send valid data then go to deep sleep
                if (!dot->getNetworkJoinStatus()) join_network();
                send_data(tx_data);
                tx_data.clear();
                dot_sleep(2);
            }
        }
```

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Data Prep during Transmission

getdatamac.py

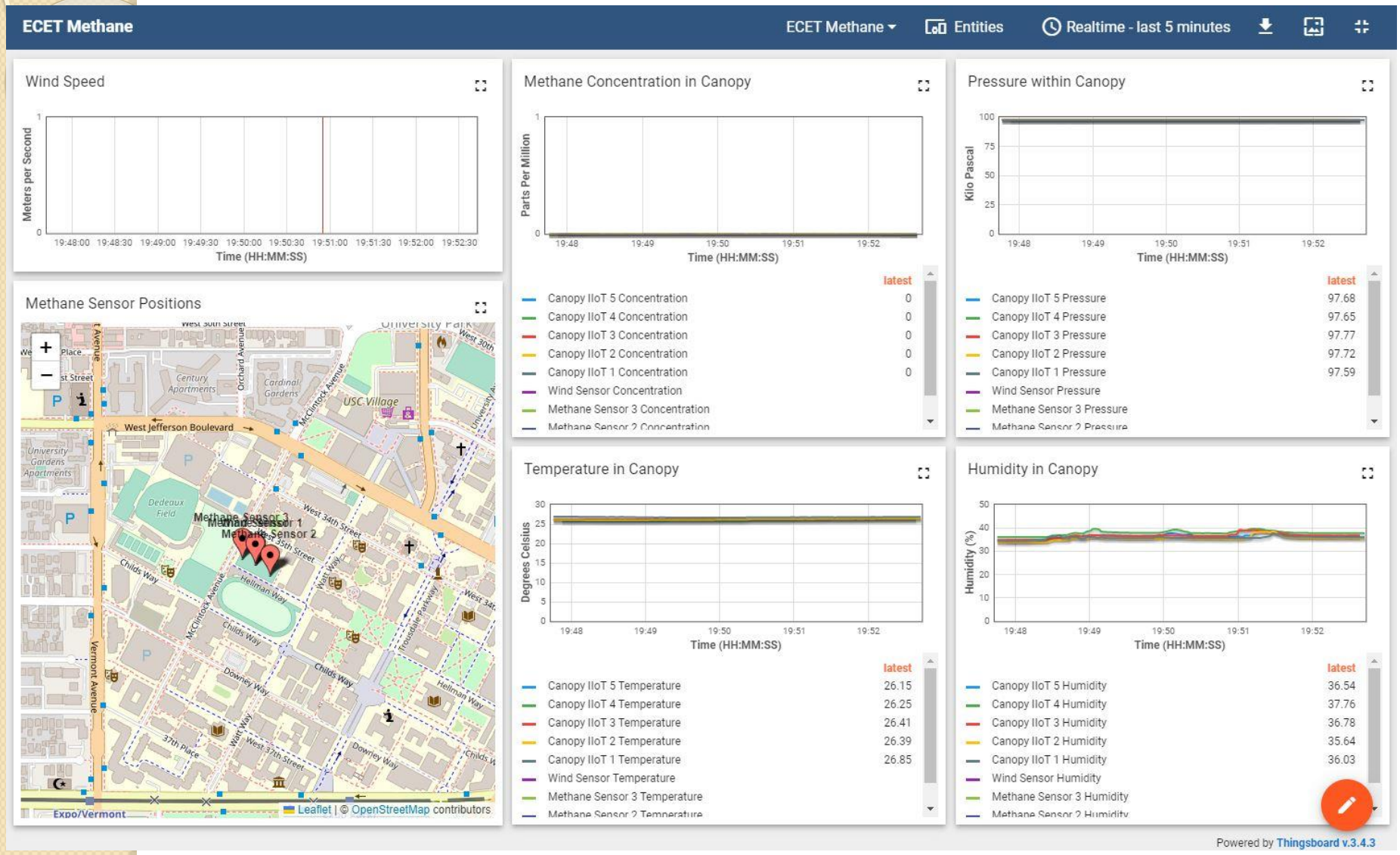
File Edit View

```
import sys
import requests
import json
import datetime
import math
import struct

#PST, tzone=0 for Linux and Mac/tzone=7 for Windows
tzone = 0

#args
year= int(sys.argv[1])
mon = int(sys.argv[2])
day = int(sys.argv[3])
hour = int(sys.argv[4])
min = 0
sec = 0

#duration in hours
dur = int(sys.argv[5])
dev = sys.argv[6]
keys = sys.argv[7]

beg_time = datetime.datetime(year, mon, day, hour, min, sec)+datetime.timedelta(hours=tzone)
end_time = beg_time+datetime.timedelta(hours=dur)
inter = int((end_time-beg_time).total_seconds()*1000)
btime = beg_time.strftime('%s')+'000'
etime = end_time.strftime('%s')+'000'

jwt = requests.post('http://frontgate.tplinkdns.com:8080/api/auth/login', headers={'Content-Type': 'application/json', 'Accept':
'application/json'}, json={'username': 'tenant@thingsboard.org', 'password': 'ecet2023'}).json()['token']

url = 'http://frontgate.tplinkdns.com:8080/api/plugins/telemetry/DEVICE/'+dev+'/values/timeseries?
```

Timestamp

Ln 1, Col 1 100% Unix (LF) UTF-8

Visualization: Real-time Acquisition

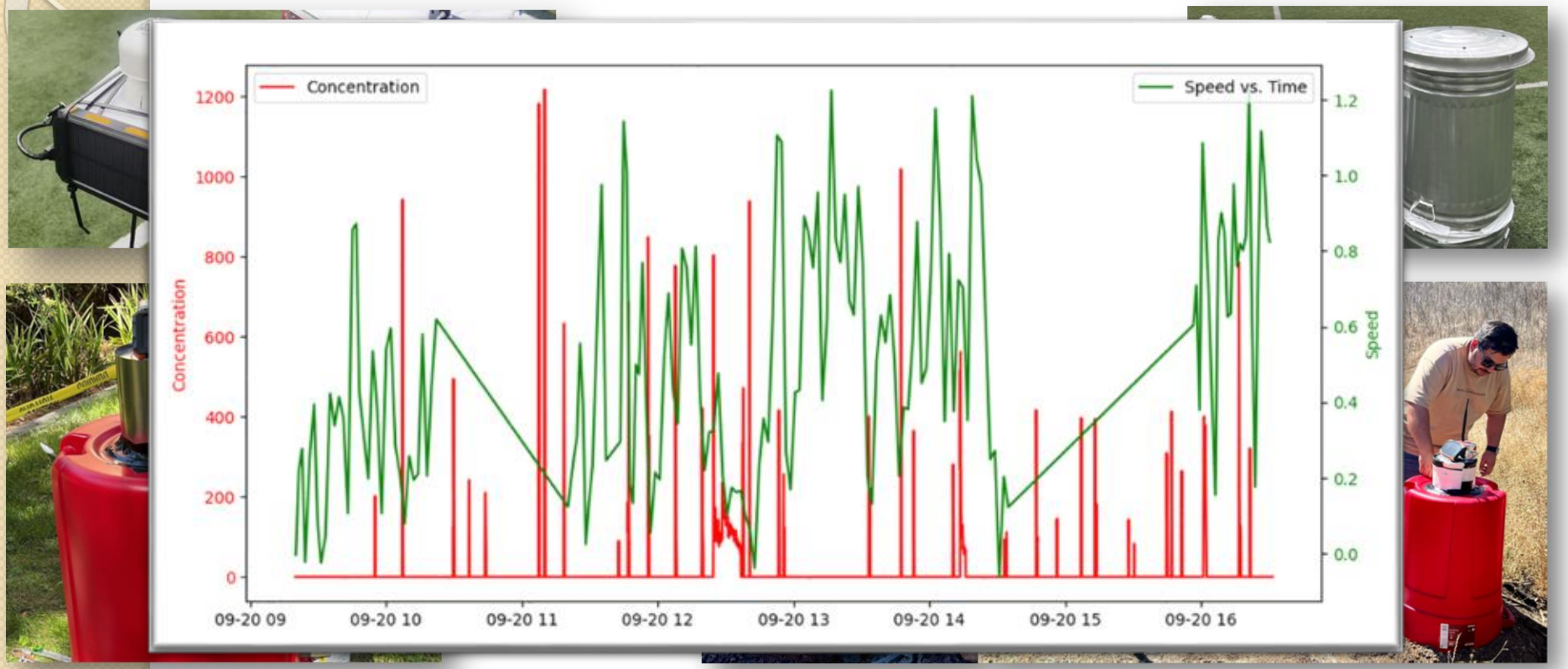


Database Access in the Cloud

getdatamac.py

File Edit View

```
dtime=0
for k, v in data.items():
   entry.append(k)
   i=0
   if k!='time':
       for sv in v:
           i=i+1
           val=sv['value']
           if val[0]=='x':
              val=val[1:]
              val=struct.unpack('!f', bytes.fromhex(val))[0]
           enval.append(val)
   else:
       for sv in v:
           if dtset==0:
               dtime=int(sv['ts'])-int(sv['value'])
               dtset=1
           i=i+1
           val=int(sv['value'])+dtime
           val=((val/3600000)-7)/24+25569
           enval.append(val)
   ennum.append(i)


nument = len(entry)-1
if nument>=0:
    for x in range(nument):
        print(entry[x]+',',end ="")
    print(entry[nument])
    numidx = ennum[0]
    for y in range(numidx):
        for x in range(nument):
            print(enval[x*numidx+y],',',end ="")
        print(enval[nument*numidx+y])
```

Ln 1, Col 1 | 100% | Unix (LF) | UTF-8

Uncovering Real World Problem